



K. J. Somaiya School of Engineering

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Experiment / assignment / tutorial No. 2

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date 27/01/24

TITLE: Effect of coupling capacitor on CE Amplifier parameters

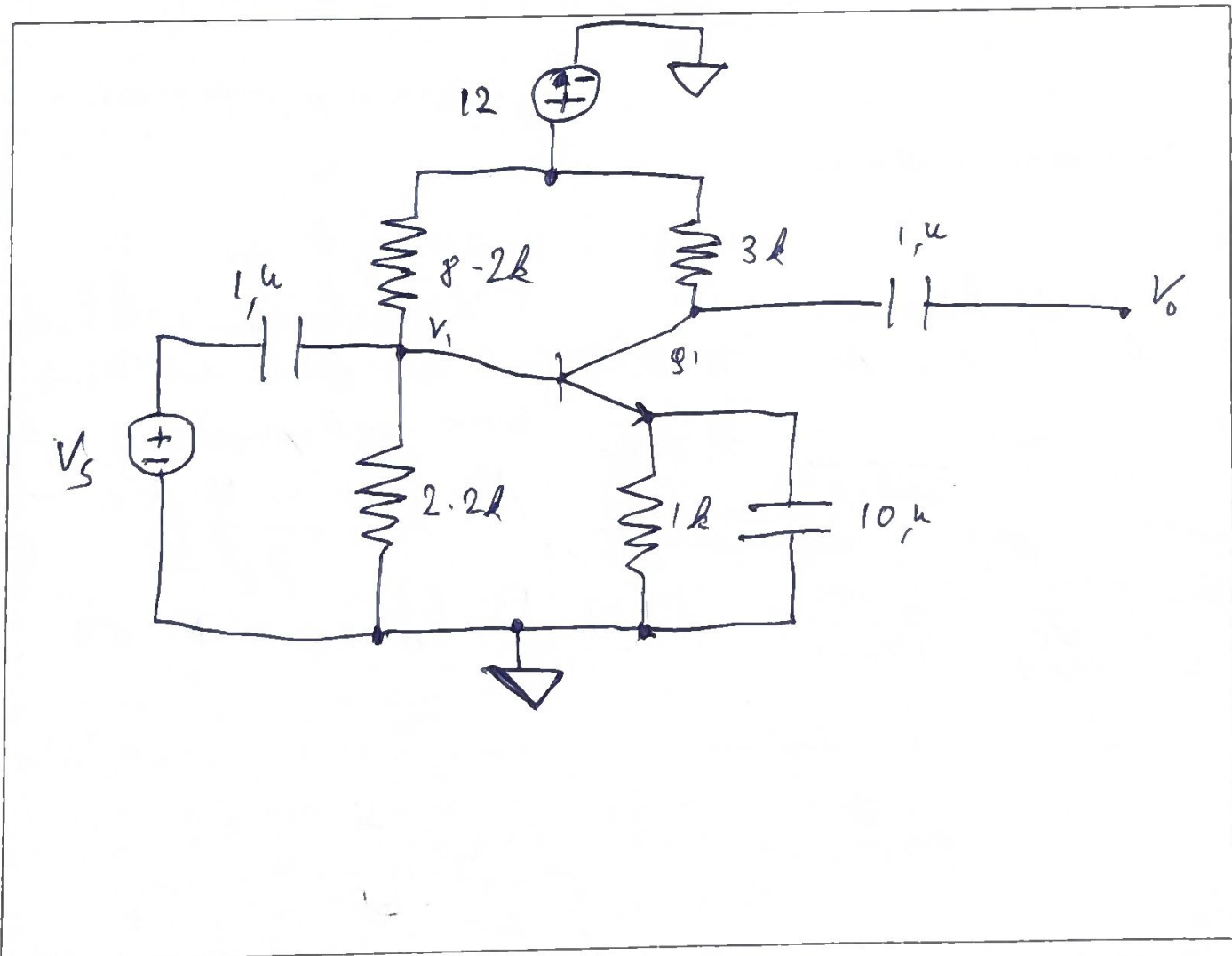
AIM: To study effect of bypass capacitor on bandwidth and gain of single stage CE amplifier.

OUTCOME: Students will be able to analyse the frequency response of BJT amplifier

PROCEDURE:

1. Implement the circuit as per the circuit diagram.
2. Take sinusoidal input signal of fixed amplitude and apply it to the amplifier circuit.
3. Vary the frequency of the source and measure the output voltage.
4. Find frequency response of amplifier circuits with reference to observation table I.
5. Plot frequency response and find lower cut-off frequency, upper cut-off frequency, voltage gain.
6. Connect at least two different bypass capacitors and repeat above process.

Circuit Diagram:





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Observation Table

$V_{CC} = 12 \text{ V}$

Input Signal Voltage (V_s) = 20 mV

	Lower cut off frequency		Upper cut off frequency		O/P V_o	Voltage Gain (V_o/V_s)		Bandwidth
	Th	Pr	Th	Pr		Th	Pr	
Without bypass Capacitor	92.5	106.4	-	30.196	58.8	3	2.94	30-196
With bypass capacitor = 2	1.13k	1.14k	-	29.01	3.78	210	189	29.00
With bypass capacitor = ?	132	180	-	29.76	3.78	210	189	29.76
	132	130		30.3	3.78	210	189	30.30

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Theoretical Calculations

Lower cut-off frequency, higher cut-off Frequency, mid band voltage gain

Voltage Gains
→ Without bypass capacitor (CE)

$$A_v = \frac{V_o}{V_s} = \frac{58.8 \times 10^{-3}}{20 \times 10^{-3}} = 2.94$$

with bypass capacitor (C_b)

$$A_v = \frac{V_o}{V_s} = \frac{3.78}{20 \times 10^{-3}} = 189$$

Conclusion:

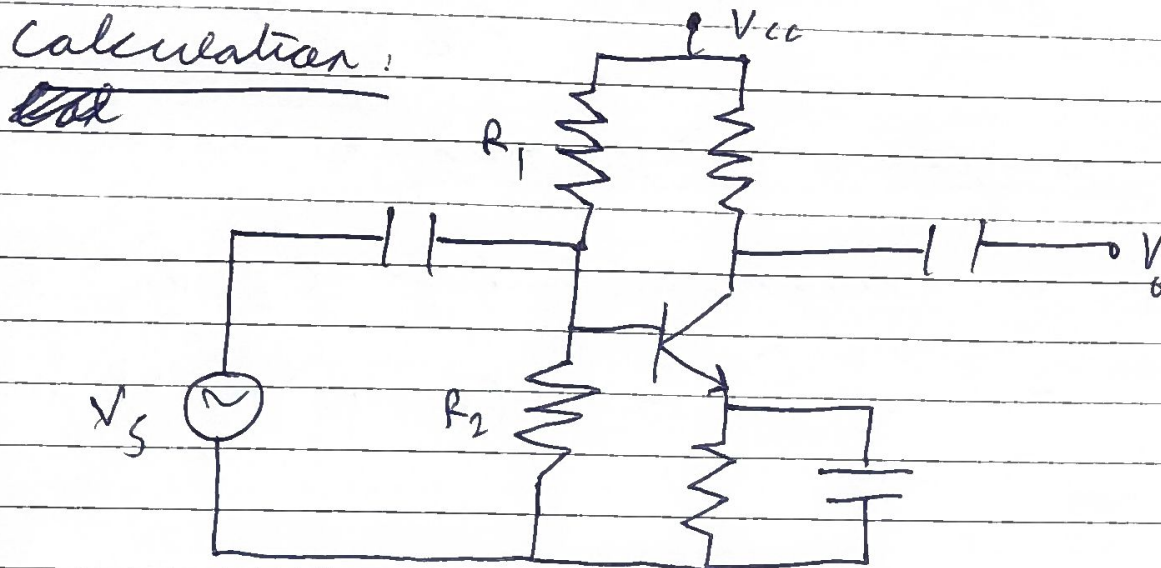
Theoretical Voltage Gain for without CE
For unbypass CE Amp
 $A_v = -\frac{(R_c || R_L)}{R_E} = \frac{-3}{1}$

$$(A_v)_{mid} = 3$$



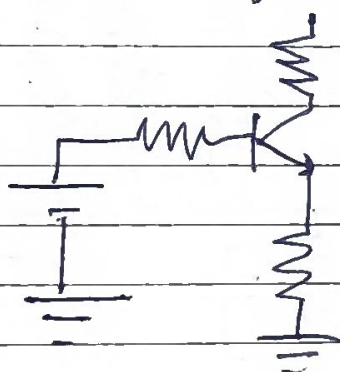
AE Experiment 2

Calculation:



(A) DC Analysis

DC Equivalent circuit, thevenins eq. circuit
KVL on input side of circuit



$$I_B = \frac{V_{TH} - V_{BE}}{R_{TH} + (\beta + 1)R_E}$$

$$= \frac{2.54 - 0.7}{(1.73)(10^3) + (271)(10^2)} = 6.75 \mu A$$

thus,

here,

$$\begin{aligned} I_C &= \beta I_B \\ &= (270)(6.75)(10^{-6}) \\ &= 1.82 \text{ mA} \end{aligned}$$

$$V_{TH} = V_{CC} \frac{R_2}{R_1 + R_2} = 2.54 \text{ V}$$

$$R_{TH} = \frac{R_1 R_2}{R_1 + R_2} = 1.73 \text{ k}\Omega$$

We are given,

$$C_u = 1.5 \text{ nF}$$

$$C_F = 11 \text{ nF}$$

(B) Small Signal Parameters :-

At room temperature, $V_T = 26 \text{ mV}$

$$g_m = \frac{I_C}{V_T} = \frac{1.82 \times 10^{-3}}{26 \times 10^{-3}} = 70 \text{ mS}$$

$$r_{\pi} = \frac{\beta}{g_m} = 386 \Omega, \quad r_c = \frac{V_T}{I_c} = 14.3 \Omega$$

$$r_o = \infty$$

(C) Mid frequency Voltage gain :-

R_E is bypassed by C_E in 3 conditions

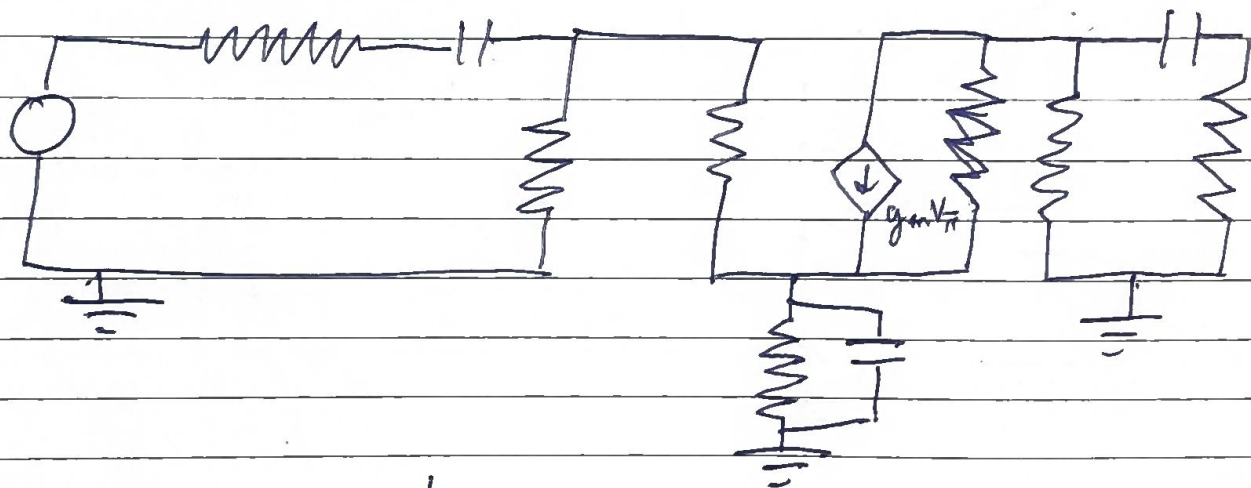
$$A_v = -g_m R_c$$

$$= -(70)(10^{-3})(3)(10^3)$$

$$|A_v|_{mid} = 210$$

(D) Low frequency Analysis :-

Hybrid π Model



$$f_{LCC1} = \frac{1}{2\pi [R_{si} + (R_{\pi} || r_{\pi})] C_{C1}}$$

$$f_{LCC2} = \frac{1}{2\pi [R_c + R_L] C_{C2}}$$

$$f_{LCE} = \frac{1}{2\pi [R_E || (\frac{R_{si} || R_c || R_L}{1+\beta}) + R_E] C_E}$$



→ When $C_E = 10 \mu F$

$$\begin{aligned} f_{LCC1} &= 132 \text{ Hz} \\ f_{LCC2} &= 53 \text{ Hz} \\ f_{LCE} &= 1.13 \text{ kHz} \end{aligned}$$

→ When $C_E = 100 \mu F$

$$\begin{aligned} f_{LCC1} &= 132 \text{ Hz} \\ f_{LCC2} &= 53 \text{ Hz} \\ f_{LCE} &= 113 \text{ Hz} \end{aligned}$$

→ When $C_E = 1000 \mu F$

$$\begin{aligned} f_{LCC1} &= 132 \text{ Hz} \\ f_{LCC2} &= 53 \text{ Hz} \\ f_{LCE} &= 11.3 \text{ Hz} \end{aligned}$$

Final Observation

When C_E is not connected

$$R_{eq} = R_{cc} + [R_1 || R_2 || R_{cc} + (1 + \beta) R_E]$$

$$= [173 \Omega || 274.4 \Omega]$$

$$= 17.19 \Omega$$

Now,

$$f_{LCC1} = \frac{1}{2\pi R_{eq} C_{C1}} = 92.5 \text{ Hz}$$

$$= 53 \text{ Hz}$$

$$f_{LCC2} = \frac{1}{2\pi (R_C + R_L) C_{C2}}$$

$$\text{Thus } f_c = 92.5 \text{ Hz}$$